

SLang

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The command language for the simulator is called **slang**. Here is the syntax and semantics (recovered from Damian’s thesis, updated by Keith):

1. *program* $\rightarrow$ *command-list*
2. *command-list* $\rightarrow$ *command-list* *command* | *command*
3. *command* $\rightarrow$ *exit* | *hi* | *city* | *edge* | *agent* | *rmedge* | *run* | *show* | *safety* | *node*
4. *exit* $\rightarrow$ **exit** | **quit**
Quits the program.

5. *hi* $\rightarrow$ **hi**

Prints “Howdy!” as a test that the program is “awake” and “listening” to command input.

6. *city* $\rightarrow$ **city** **STRING** **INT** **INT**

Creates a city object with:

- the title given by the string
- fixed number, N , of nodes numbered 1 to N
- maximum out degree for any of the nodes so pointers the edges adjacent to the node can be allocated once and have to manage variable length lists in this speed critical function.

7. *edge* $\rightarrow$ **edge** **INT** **INT** **FLOAT** **FLOAT** **FLOAT** **FLOAT** **FLOAT**

This adds an edge in the current city. The edge is identified by (from node, to node, order of entry). The time spent by a car on an edge is determined by the parameters given for each edge. The time is:

$$\text{time} = \mathcal{F} \left(1 + b \left(\frac{c}{c_{max}} \right)^\beta \right)$$

where c is the amount of traffic on the edge, c_{max} is the capacity, $\mathcal{F}$ is the free flow time, b and β are constants. The parameters in order are the following. If parameters are not given it defaults to the last used value for that positional parameter (not true for updated version - parameters must be present in SLANG file).

- From node number

- To node number
- c_{max} : Capacity as the number of cars per unit time for the road. Used in the link cost formulas.
- $\mathcal{F}$: Free flow travel time, the amount of unit time that it takes a car to travel the length of the edge under ideal conditions
- b : coefficient in the link cost formulas, also commonly named α
- β : power or exponent in the link cost formulas
- The last value specified is the safety value of the specified edge, typically ranging from 0 – 1.

8. *rmedge* → **rmedge** INT INT |
rmedge INT INT INT

Deletes the edge identified by the from and to node numbers or the from and to node numbers plus the order in which they were added.

9. *node* → **node** INT INT FLOAT FLOAT FLOAT FLOAT

Specifies node parameters for the node indicated with the following parameters:

- Node ID
- Node capacity
- Agent wait time for node - the time that an agent will wait at the node before being re-evaluated (in unit time)
- The safety of the given node
- The node longitude
- The node latitude

10. *agent* → **agent** INT INT INT

Specifies simulation agents with the following parameters:

- The starting node of the agent in each simulation (evaluation).
- The number of agents to start at the specified node.
- The number of individuals in the agent group.

11. *probabilities* → **probabilities** STRING

STRING specifies a file to read for a population of probability sets that should be used to initialize the population before running a new evolution.

12. *anyint* → **INT** | *

13. *run* → **run** INT INT INT

Initialize the simulation and run the simulation and learning. The parameters are:

- Total number of evolution generations.
- The number of time steps between reporting progress. Must evenly divide simulation steps.
- The number of runs of the whole optimization, restarting each time.

14. *simulate* → **simulate** **STRING**

STRING specifies a file with probabilities that should be run through the traffic simulation. This is used to evaluate fitness of a human-written set of probabilities. Use for testing simple cases to prove the fitness function works, and to prove the model can solve simple problems.

15. *slang* → **slang** **STRING**

Command to output a SLang file based on the current city topology. STRING designates the filename to write the SLang file to.

16. *show* → *edges* | *topology* | *probabilities* | *drawable* | *parameters*

Output commands show edges, city topology, and probabilities. The drawable command outputs the city topology information needed to create a visual representation of the graph. Parameters refer to the evolution parameters (generations, sigma, population size, etc)

17. *show edges* → **show edges**

18. *show topology* → **show topology**

19. *show probabilities* → **show probabilities**

20. *show drawable* → **show drawable**

NOTE: The grammar rules here indicate the theoretical requirements of the grammar. Rules may need to change depending on implementation and optimization algorithm. SLang was created with Ant-Colony Optimization in mind, and has been repurposed here for use with an Evolution Strategy used to optimize a cloud of traffic-assignment probabilities.